Differential forms

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Q. Given an open subset $U \subseteq \mathbb{R}^n$ and $k \leq n$, what "structure" $X_{U,k}$ can I have on U such that for *every* k-manifold M inside U, there is a "nice" notion of taking "integral of $X_{U,k}$" over M?

A. Suppose that to every point $p \in U$, we have assigned a function φ_p which takes in k-parallelepipeds, and converts them into real numbers.

Now, given any k-manifold M, we can split it into many infinitesimal pieces which look like k-parallelepipeds, and then φ_p will tell us what value to assign to each parallelepiped. Finally, we take the sum over all of these values to compute the integral over M.

Note that we would prefer if φ_p was "coherent" in assigning values to the k-parallelepipeds, for example, if I scale a parallelepiped A to get A', then $\varphi_p(A')$ should equal $2\varphi_p(A)$.

This matches with the intuition we have developed in the section **Building intuition about exterior powers**. We want to have each $\varphi_p \in \Lambda^k(\mathbb{R}^n)^*$.

This "structure" that we have developed, is called a **differential k-form over U**.

Given an open subset $U \subseteq \mathbb{R}^n$, a **differential k-form over U** is a function ω of type

$$\omega :: (p : U) \to \left(\Lambda^k T_p(U)\right)^* \cong \Lambda^k T_p^*(U)$$

We shall write "k-form" as short-hand for "differential k-form".